INSURANCE RISK & CLAIMS ANALYSIS

BUSINESS REQUIREMENT

An insurance company is looking to better understand its **policyholder base and claim patterns** to make data-driven business decisions. Currently, policy and claims data are scattered across multiple sources, making it difficult for stakeholders to track performance and identify trends.

The company requires a **centralized interactive dashboard** in Power BI that can provide a clear overview of:

KPI's Requirements:

1. **Total Policies** – to measure the size of the active customer base.
2. **Total Claim Amount** – to track the overall financial impact of claims.
3. **Claim Frequency** – to analyse how often claims are being made.
4. **Average Claim Amount** – to assess claim severity and potential risk exposure.
5. **Gender-wise Total Policies** – to understand customer distribution across genders for better segmentation and policy targeting.

Chart's Requirements:

To **deep dive into the data**, we need to go beyond KPIs and analyse different aspects of insurance policies and claims. Charts help us **visually explore patterns, relationships, and anomalies** across customer demographics, car details, and claim behaviours. By analysing charts, stakeholders can **identify risk factors, understand customer segments, and optimize policy decisions**.

For this report, all visualizations are designed around **two key dynamic measures**:

- **Total Claim Amount**
- **Total Policies**

These measures provide the foundation to compare, filter, and segment the data effectively.

Visualization Requirements:

1. **By Car Use (Donut Chart)** – To analyse policy distribution and claim amounts based on how cars are being used (e.g., personal, commercial).
2. **By Car Make (Bar Chart)** – To identify which car brands have higher policies and claims, highlighting brand-based risks.
3. **By Coverage Zone (Donut Chart)** – To evaluate policies and claims by geographic zones, useful for regional risk analysis.
4. **By Age Group (Frequency Chart/Histogram)** – To assess policyholders' age distribution and identify which age brackets file more claims.

5. **By Car Year (Area Chart)** – To analyse how the car's age (year of manufacture) impacts policy counts and claim amounts.
6. **By Kids Driving (Ribbon Chart)** – To compare the impact of young drivers in households on policy count and claim amounts.
7. **By Education (Pie Chart)** – To understand how education levels correlate with insurance policy adoption and claims.
8. **By Education & Marital Status (Matrix Heat Grid)** – To explore the combined effect of education and marital status on policies and claims, highlighting customer profiles.

Learning Source

This dashboard was built as part of my Power BI practice using a dataset from a YouTube tutorial. I recreated and customized the dashboard to strengthen my understanding of data modeling, DAX measures, Power Query transformations, and KPI design.